

SCm Qubit T_2 Coherence via $F_{U,Bi}/F_U$ Ratio and Thermal Protection

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Abstract

We derive the Star Cradle Mechanics (SCm) qubit coherence time T_2 formula, which incorporates the buoyancy field ratio $F_{U,Bi}/F_U$ and an exponential thermal protection factor from the SCm gap energy Δ_{SCm} :

$$T_2 = \frac{\hbar}{\Delta_{\text{SCm}}} \cdot \exp\left(\frac{\Delta_{\text{SCm}}}{k_B T}\right) \cdot S_{26}^{(3)} \cdot \frac{F_{U,Bi}}{F_U}$$

This extends conventional qubit T_2 models by encoding the UQFF buoyancy protection mechanism and the 26-dimensional phonon coupling.

1. Introduction

In superconducting and topological qubit architectures, the transverse relaxation time T_2 governs the duration over which quantum phase coherence is maintained. Standard models relate T_2 to the energy gap Δ and temperature T via:

$$T_2^{\text{std}} = \frac{\hbar}{\Delta} \cdot \exp\left(\frac{\Delta}{k_B T}\right)$$

The SCm framework introduces two additional factors:

1. The **26-dimensional buoyancy prefactor** $S_{26}^{(3)}([\text{SSq}]) = (1 - [\text{SSq}])^3$
2. The **buoyancy field ratio** $F_{U,Bi}/F_U$ capturing the fraction of the unified field contributing to topological protection

2. SCm Gap Energy

The SCm gap energy is derived from the characteristic 1.25 THz phonon mode:

$$\Delta_{\text{SCm}} = \hbar \omega_{\text{phonon}} \cdot 2\pi = \hbar \cdot 2\pi \cdot 1.25 \times 10^{12} \text{ Hz}$$

$$\Delta_{\text{SCm}} \approx 8.27 \times 10^{-22} \text{ J} \approx 5.17 \text{ meV}$$

3. The SCm T_2 Formula

$$T_2^{\text{SCm}} = \underbrace{\frac{\hbar}{\Delta_{\text{SCm}}}}_{\text{base time}} \cdot \underbrace{\exp\left(\frac{\Delta_{\text{SCm}}}{k_B T}\right)}_{\text{thermal suppression}} \cdot \underbrace{S_{26}^{(3)}}_{\text{26D buoyancy}} \cdot \underbrace{\frac{F_{U,Bi}}{F_U}}_{\text{buoyancy ratio}}$$

Parameter Regimes

Regime	T (K)	$\exp(\Delta/k_B T)$	Application
Dilution fridge	0.015	$\sim 10^{174}$	Superconducting qubits
^3He cryostat	0.3	$\sim 10^8$	Topological qubits
Liquid He	4.2	~ 14.4	High- T_c devices

Enhancement Factor

The SCm enhancement over standard T_2 :

$$\eta_{\text{SCm}} = S_{26}^{(3)} \cdot \frac{F_{U,Bi}}{F_U}$$

For $[\text{SSq}] = 0.57$ and $F_{U,Bi}/F_U = 0.3$:

$$\eta_{\text{SCm}} = 0.0795 \times 0.3 = 0.0239$$

This represents a buoyancy-mediated coherence channel distinct from conventional T_2 decay pathways.

4. Physical Interpretation

The $F_{U,Bi}/F_U$ ratio encodes the fraction of the unified buoyancy field available for topological qubit protection. When the buoyancy component dominates ($F_{U,Bi}/F_U \rightarrow 1$), the qubit is maximally protected by the phonon-buoyancy shield. When $F_{U,Bi} \ll F_U$, other force components (magnetic, gravitational) dominate and coherence protection diminishes.

5. Implementation

Calculator: `SCmQubitT2CoherenceFUBiRatioCalculator` in `Condensed-Physics.py`

- Accepts temperature T , field strengths F_U and $F_{U,Bi}$, phonon frequency, and $[\text{SSq}]$
- Outputs T_2^{SCm} , T_2^{std} , enhancement factor, and thermal exponential

6. Conclusion

The SCm T_2 formula provides a first-principles connection between UQFF buoyancy physics and qubit decoherence, predicting coherence modulation through the $F_{U,Bi}/F_U$ ratio and the 26-dimensional phonon coupling.

References

- Koch, J. et al. (2007). Charge-insensitive qubit design from Cooper pair box. *Phys. Rev. A* **76**, 042319.
- Murphy, D.T. (2025). UQFF: Star Cradle Mechanics framework. Star-Magic repository.
- PAPER_1098: Phonon-Mediated Qubit Gate Fidelity.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. *Phys. Rev. Lett.* **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
4. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. *A&A* **356**, 788 — arXiv:astro-ph/0004212
5. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. *MNRAS* **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
6. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics*. Harcourt
7. Kittel, C. (2004). *Introduction to Solid State Physics*. 8th ed. Wiley
8. Feynman, R.P. (1982). *Simulating Physics with Computers*. *Int. J. Theor. Phys.* **21**, 467 — doi:10.1007/BF02650179
9. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. *Stud. Hist. Phil. Mod. Phys.* **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
10. Weinberg, S. (1989). *The Cosmological Constant Problem*. *Rev. Mod. Phys.* **61**, 1 — doi:10.1103/RevModPhys.61.1